

Differentiable functions – 2

Mean value theorems: how to produce a point ξ

Homework

1 сентября 2026 года

Theoretical minimum. Rolle, Lagrange and Cauchy mean value theorems; the auxiliary function trick and the standard multipliers $e^{\lambda x}$, $\frac{1}{x}$, $\varphi(x)$; Darboux' theorem; repeated application of Rolle to a function with many zeros; the mean value theorem for integrals and its use in producing a point.

1. (IMC 2025) Let $f: (0, \infty) \rightarrow \mathbb{R}$ be continuously differentiable and let $b > a > 0$ satisfy $f(a) = f(b) = k$. Prove that there is $\xi \in (a, b)$ with $f(\xi) - \xi f'(\xi) = k$.
2. (RMO 1999) Let $a < b$ and let $f, g: [a, b] \rightarrow \mathbb{R}$ be differentiable with increasing derivatives and $f'(a) > 0$, $g'(a) > 0$. Prove that there is $c \in [a, b]$ with $\frac{f(b)-f(a)}{b-a} \cdot \frac{g(b)-g(a)}{b-a} = f'(c)g'(c)$.
3. (OMOUS 2022) Let $f: [0, 1] \rightarrow \mathbb{R}$ be continuous with $\int_0^1 f(x) dx = \frac{\pi}{4}$. Prove that there is $x_0 \in (0, 1)$ with $\frac{1}{1+x_0} < f(x_0) < \frac{1}{2x_0}$.
4. Let f be continuous on $[a, b]$ and differentiable on (a, b) , where $b - a \geq \pi$. Prove that there is $x_0 \in (a, b)$ with $f'(x_0) < 1 + f(x_0)^2$.
5. (IMC 2016) Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Suppose f has infinitely many zeros but there is no $x \in (a, b)$ with $f(x) = f'(x) = 0$.
 - a) Prove that $f(a)f(b) = 0$.
 - b) Give an example of such a function on $[0, 1]$.
6. (IMC 2005) Prove that for a three times differentiable $f: \mathbb{R} \rightarrow \mathbb{R}$ there is $\xi \in (-1, 1)$ with $\frac{f'''(\xi)}{6} = \frac{f(1)-f(-1)}{2} - f'(0)$.

7. Let $f \in C^1[a, b]$ with $f' > 0$. Prove that $\frac{f(b)-f(a)}{b-a} \leq \max_{x \in [a, b]} \left(f'(x) + \frac{(f'(x))^2}{4f(x)} \right)$.
8. Let f be continuous on $[a, b]$ and differentiable on (a, b) , where $0 < a < b$. Prove that there are $c_1, c_2 \in (a, b)$ with $\frac{f'(c_1)}{a+b} = \frac{f'(c_2)}{2c_1}$.
9. (IMC 1995) Let f be twice continuously differentiable on $(0, +\infty)$ with $\lim_{x \rightarrow 0^+} f'(x) = -\infty$ and $\lim_{x \rightarrow 0^+} f''(x) = +\infty$. Show that $\lim_{x \rightarrow 0^+} \frac{f(x)}{f'(x)} = 0$.
10. (IMC 2006) Find all sequences $a_0, a_1, \dots, a_n$ of reals with $n \geq 1$ and $a_n \neq 0$ for which the following holds: if $f: \mathbb{R} \rightarrow \mathbb{R}$ is n times differentiable and $x_0 < x_1 < \dots < x_n$ satisfy $f(x_0) = f(x_1) = \dots = f(x_n) = 0$, then there is $h \in (x_0, x_n)$ with $a_0 f(h) + a_1 f'(h) + \dots + a_n f^{(n)}(h) = 0$.